Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Universidad de Valladolid
Short name / acronym used in projects	UVA
Participant Identification Code (PIC), if available	If no PIC is available yet, please indicate 'not yet available'.
Website(s)	https://www.ibgm.uva.es/investigaci%C3%B3n/grupos-de-investigaci%C3%B3n/calcio-y-funci%C3%B3n-celular
Country	Spain
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Sanz y Forés 3, 47003 Valladolid, Spain
Department(s), institute(s), research group(s) or unit(s) that would be involved	Calcium and Cellular Function Group, Institute for Biomedicine and Molecular Genetics (IBGM)
Additional institutional resources relevant to collaboration	Facilities and equipment's of the IBGM-UVa (https://www.ibgm.uva.es/*and* centralized services of UVa (https://aplicaciones.educa.jcyl.es/infrared/welcome.action).

Main contact person for this profile	Maria Teresa Alonso, Full professor, Departemnt of Biochemistry, Molecular Biology and Physiology
Main contact email	talonso@uva.es

Other contact person(s), if relevant	Name, role, email

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	X Other (please specify): Advanced imaging technologies

Free Keywords for internal searchable database (please provide up to 10)	calcium imaging; live-cell imaging; Ca2+ biosensors; luminescence; organelle crosstalk; aequorin; transgenics; aging; neurodegeneration; Drosophila and mouse models.
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
Our research has focused on the role of intracellular calcium as a signaling molecule in various cellular paradigms and models, both in health and disease. Our group was a pioneer in Spain in microfluorescence and Ca2+ imaging in living cells, with single cell resolution, generating indicators, protocols, infrastructures and methodologies for monitoring Ca2+ (know-how) that has gained us an international reputation in the calcium signaling field. In the last two decades our group has focused on the role of the cytoplasmic organelles in Ca2+ signaling, especially mitochondria, endoplasmic reticulum and nucleus, and recently the endo-lysosome. To address this, we have developed a number of novel tools based on the bioluminescent Ca2+ binding protein aequorin, which have provided highly novel insights in the Ca2+ signaling field. For instance, we have demonstrated that chromaffin cells exhibit a very specialized calcium mechanism (CICR) typical of heart cells (JCB, 1999) and reported for the first time that mitochondria can take up unexpectedly high amounts of Ca2+ in the millimolar range under physiological conditions (Nat Cell Biol, 2000) or reported mitochondrial (JBC, 2001) and nuclear calcium oscillations (JBC, 2002). We have contributed to the role of organelles in neurosecretory cells (chromaffin and pituitary) as well as primary cerebellar granule cells. We have also generated a new family of fluorescent calcium genetic indicators optimized for organelles (PNAS, 2014; Cell Chem Biol, 2016). Our research extends from a molecular level in vitro, using the purified protein to explore mechanistic aspects of the structure-function relationship, up to level of the complete organism, with the generation of transgenic animals and their application in in vivo studies in the whole animal in physio-pathological aspects such as aging (JCS, 2020) or neurodegeneration in models such as Huntington´s disease.

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

Because Intracellular calcium acts as a ubiquitous second messenger across almost all living cells, measuring intracellular Ca2+ dynamics provides a direct readout for several distinct processes. Ca2+ signalling is a primary proxy for cellular activity, metabolic demand and cellular excitability. It can be widely applied in very different systems or models such as drug screening, live-cell functional analysis and neurodegenerative disease modelling. Therefore, the specific methodologies provided by the group could support translational eye-health projects focused on retinal neurodegeneration, cellular stress signalling and biomarker discovery.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- *Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.*
- live-cell calcium imaging
- Intraorganellar calcium imaging
- biosensor development
- image analysis
- microfluorescence
- intracellular signaling
- luminal endo-lysosomal calcium dynamics
- transgenic Drosophila and mouse disease models
- neurodegeneration research

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

Advanced microscopy and microfluorescence platforms; luminescent equipments; genetically encoded calcium sensors; live-cell imaging systems; experimental transgenic Drosophila and mouse models;

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

Training in calcium imaging methodologies, generation of Drosophila lines, in vivo imaging of intraorganellar calcium, biosensor use, image analysis, functional live-cell assays and intracellular signalling studies

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

Access to advanced retinal disease models, collaborations in translational neuroscience and ophthalmology, integration of calcium signalling approaches into retinal and neurodegenerative disease research, application of the calcium technologies to the ophthalmology field.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
María Teresa Alonso	Full Professor	Biochemistry	Researcher	Experienced	talonso@uva.es
Thomas Schimmang	National Research Council (CSIC) Investigator	Developmental Biology	Researcher	Experienced	thomas.schimmang@uva.es
Jonathan Rojo	Senior Researcher	Nursing	Researcher	Experienced	jonathan.rojo@uva.es
Leonardo Gallegos	Postdoctoral Researcher	Biochemistry	Researcher	Experienced	leonardo.gallegos@uva.es
Beatriz Callejo	PhD student	Biochemistry	Researcher	Early stage	beatriz.callejo@uva.es
Cynthia Sánchez	PhD student	Biochemistry	Researcher	Early stage	cynthia.sanchez@uva.es

Main disciplines represented in your team	physiology, molecular biology, cell biology, genetics,
Experience in EU / international / multi-partner collaborative projects	M.T.A. participated as a researcher in the project "Neuron -Disturbance of calcium processing by intracellular organelles in the ER-mitochondria calcium cycle in a triple mutant model of Alzheimer disease." Ref.: ERA-net NEURON - SAF2008-03175-E. Funding Agency: European Commission/MICINN. Position: P.I. Spanish partner, University of Valladolid (J. García-Sancho). Start: 2009 – End: 2012. Please indicate the most relevant experience in 1 or 2 lines (no specific details here)
Experience with clinical, translational or patient-facing research (if relevant)	X Yes ☐ No ☐ Not applicable | Short note: Experience with stem cell models, viral gene transfer, CRISPR/Cas9-mediated gene modification

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

1. Calvo, B., Torres-Vidal, P., Delrio-Lorenzo, A., Rodriguez, C., Aulestia, F.J., Rojo-Ruiz, J., Callejo, B., McVeigh, B.M., Keller, M., Grimm, C., Oorschot, V., Moiseenkova-Bell, V., Yule, D.I., Garcia-Sancho, J., Patel, S. and Alonso, MT*. (2025). Direct measurements of luminal Ca2+ with endo-lysosomal GFP-aequorin (ELGA) reveal functional IP3 receptors. J.Cell Biol. 224. We provide a novel tool to measure luminal endo-lysosomal Ca2+ and novel insights into the mechanisms governing Ca2+ handling in the endo-lysosomal system. This work has been selected for the Spotlight section in the same issue (Morgan, AJ (2025) J Cell Biol 224). Also, this work is the main body of P. Torres´s PhD thesis.

2. Delrio-Lorenzo, A., Rojo-Ruiz, J., Torres-Vidal, P., Alonso, M.T*., and García-Sancho, J. (2024). In vitro and in vivo calibration of low affinity genetic Ca2+ indicators. Cell Calcium 117, 102819. We described here a novel calibration procedure to accurately convert the signal of the fluorescent indicator in high Ca2+, like the endoplasmic reticulum (ER) into [Ca2+]ER, both in vitro and in vivo (in erGAP3 transgenic flies). This work is part of the PhD thesis of A. Del Río.

3. Gómez-Valadés AG, Pozo M, Varela L, … Alonso MT (18/26), …, Claret M. (2021) Mitochondrial cristae-remodeling protein OPA1 in POMC neurons couples Ca2+ homeostasis with adipose tissue lipolysis. Cell Metab. 33: 1820-1835. Dr. Claret contacted us as experts in mitocondrial Ca2+ to image mitocondrial Ca2+ specifically in POMC neurons in acute hypothalamus slices of KO-OPA1 mice.

4. Rojo-Ruiz, J., Navas-Navarro, P., Nuñez, L., García-Sancho, J., and Alonso, MT*. (2020) Imaging of Endoplasmic Reticulum Ca2+ in the Intact Pituitary Gland of Transgenic Mice Expressing a Low Affinity Ca2+ Indicator. Frontiers in Endocrinology 11: 615777. doi: 10.3389/fendo. 615777. In this article we showed that the erGAP3 transgenic mice show robust functional expression in pituitary gland where we have simultaneously studied the Ca2+ signals evoked by various releasing hormones in the intact gland. We demonstrated unambigously the existance of polireceptorial cells that respond to several hormones. This work was part of the P. Navas's PhD thesis.

5. Delrio-Lorenzo A, Rojo-Ruiz J, Alonso MT, García-Sancho J. (2020) Sarcoplasmic reticulum Ca2+ decreases with age and correlates with the decline in muscle function in Drosophila. J Cell Sci. 133: jcs240879. In this work we tested the hypothesis that the age-associated loss of mass and muscle force (sarcopenia) has its origin in the loss of Ca2+ homeostasis in the sarcoplasmic reticulum (SR). We indeed could demonstrate that aged flies stored 10-fold less Ca2+ in the SR of thoracic muscles than young flies and this correlated well with the process of sarcopenia. This work was the main body of A. del Río's PhD thesis.

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). *Indicate if international.*

1. PID2023-146434NB-I00. Physiological and molecular basis of endo-lysosomal Ca2+ dynamics in health and disease. DGI MICINN. P.I. MTA. Start: 01/01/2025 – End: 31/8/2027. 325.000€.

2. PID2020-116086RB-I00. Calcium and Cellular Function. DGI MICINN. P.I. MTA. (Co-PI: J. García-Sancho). Start: 01/01/2022 – End: 31/8/2024. (extended until 31/12/2024) 340.000€.

3. BFU 2017-83066-P. Calcium and Cellular Function. DGI MICINN. P.I. MTA (Co-PI: J. García-Sancho). Start: 01/01/2018 – 31/12/2020 (extended until 31/12/2021). 363.000€.

4. BFU2014-53469-P. Calcio y Función Celular. DGI MICINN. Position: Co-P.I. (P.I. - J. García-Sancho). 01/01/2015 – 31/12/2017. 423.500€.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Please indicate only what can be shared at this stage.

1. P201230475. M.T. Alonso and J. García-Sancho Martín. Calcium sensors and methods for the detection of intracellular free calcium. Country of priority: Spain. 29.03.2012. University of Valladolid

2. P201231104. M.T. Alonso and J. García-Sancho Martín. Apoaequorin mutants and methods for their use. Country of priority: Spain. 13.07.2013. University of Valladolid.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers X Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	X Academic only ☐ Non-academic only ☐ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	e.g. visa, insurance, access to facilities, language, employment status, health & safety induction

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile. Calcium imaging, intraorganellar calcium measurements, live-cell biosensors, in vivo functional assays and neurodegenerative disease models

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. Advanced retinal disease models, translational ophthalmology collaborations, AI-based image analysis and multimodal neurodegeneration approaches

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X Understood

Name: Maria Teresa Alonso
Date completed: 22/9/2026

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